

## INOWROCLAW, Poland

WMO: 123420

Lat: 52.807N

Lon: 18.277E

Elev: 272

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |            |            | 99.6%                         |            |            | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>3.8 | (c)<br>8.8 | (d)<br>-0.3                   | (e)<br>5.5 | (f)<br>5.7 | (g)<br>3.8 | (h)<br>6.7 | (i)<br>10.2 | (j)<br>30.0           | (k)<br>33.1 | (l)<br>27.8 | (m)<br>33.8 | (n)<br>6.7            | (o)<br>80 | (p)<br>0.606 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>19.0               | (c)<br>88.0     | (d)<br>68.2 | (e)<br>84.5 | (f)<br>66.8 | (g)<br>81.1 | (h)<br>65.4 | (i)<br>70.3         | (j)<br>82.8 | (k)<br>68.9 | (l)<br>80.7 | (m)<br>67.3 | (n)<br>77.9 | (o)<br>9.9              | (p)<br>120 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.7                            | 99.1 | 74.1 | 65.1 | 93.7 | 72.3 | 63.4 | 88.2 | 71.4 | 34.4          | 84.0 | 33.1 | 80.3 | 31.9 | 77.5 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |    |    |
|-------------------|----------|----------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|----|----|
|                   |          |          | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     | DB | WB |
| (a) 25.6          | (b) 22.6 | (c) 20.1 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |
| (a) 25.6          | (b) 22.6 | (c) 20.1 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.6   | 27.9  | 30.4  | 38.9  | 49.4  | 57.5  | 64.1  | 67.7  | 66.9  | 57.8  | 47.4  | 40.6  | 33.7 |
|                                                                                  | DBStd       | 15.61  | 9.65  | 10.10 | 8.46  | 7.72  | 7.55  | 5.87  | 5.71  | 5.66  | 5.86  | 7.19  | 6.09  | 8.67 |
|                                                                                  | HDD50       | 2643   | 685   | 548   | 352   | 104   | 19    | 0     | 0     | 0     | 6     | 138   | 284   | 507  |
|                                                                                  | HDD65       | 6292   | 1150  | 968   | 810   | 470   | 251   | 86    | 36    | 43    | 229   | 546   | 731   | 972  |
|                                                                                  | CDD50       | 2142   | 0     | 0     | 7     | 87    | 252   | 424   | 549   | 524   | 240   | 57    | 2     | 0    |
|                                                                                  | CDD65       | 318    | 0     | 0     | 0     | 3     | 20    | 60    | 120   | 102   | 13    | 0     | 0     | 0    |
|                                                                                  | CDH74       | 3170   | 0     | 0     | 0     | 55    | 258   | 631   | 1079  | 985   | 162   | 0     | 0     | 0    |
|                                                                                  | CDH80       | 1009   | 0     | 0     | 0     | 15    | 47    | 198   | 367   | 325   | 57    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 9.8    | 11.0  | 10.6  | 10.6  | 10.5  | 9.3   | 9.0   | 8.8   | 8.4   | 9.2   | 9.5   | 10.0  | 11.1 |
| Precipitation                                                                    | PrecAvg     | 20.6   | 1.2   | 1.0   | 1.3   | 1.2   | 2.1   | 2.2   | 3.3   | 2.2   | 1.9   | 1.3   | 1.4   | 1.3  |
|                                                                                  | PrecMax     | 29.3   | 2.9   | 2.4   | 2.6   | 3.9   | 5.3   | 4.5   | 6.7   | 6.9   | 4.4   | 4.4   | 4.9   | 3.5  |
|                                                                                  | PrecMin     | 14.0   | 0.1   | 0.2   | 0.4   | 0.2   | 0.6   | 0.8   | 1.1   | 0.5   | 0.5   | 0.2   | 0.1   | 0.3  |
|                                                                                  | PrecStd     | 3.4    | 0.7   | 0.5   | 0.6   | 0.7   | 1.0   | 1.1   | 1.6   | 1.3   | 1.1   | 0.9   | 0.8   | 0.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.6  | 50.1  | 66.0  | 82.2  | 84.3  | 92.8  | 92.7  | 92.4  | 86.9  | 71.1  | 57.2  | 52.2 |
|                                                                                  |             | MCWB   | 45.7  | 44.9  | 53.2  | 61.5  | 64.1  | 69.1  | 69.3  | 69.8  | 66.6  | 60.5  | 54.2  | 48.9 |
|                                                                                  | 2%          | DB     | 46.0  | 47.6  | 59.0  | 73.0  | 80.5  | 85.2  | 87.9  | 87.3  | 77.8  | 65.5  | 53.3  | 48.7 |
|                                                                                  |             | MCWB   | 43.5  | 43.9  | 48.7  | 56.8  | 62.7  | 67.9  | 68.6  | 67.9  | 63.3  | 57.8  | 50.5  | 46.7 |
|                                                                                  | 5%          | DB     | 42.9  | 44.8  | 54.8  | 68.1  | 76.6  | 80.7  | 84.1  | 83.8  | 72.6  | 61.7  | 51.1  | 46.6 |
|                                                                                  |             | MCWB   | 41.2  | 41.6  | 47.5  | 54.5  | 61.1  | 65.2  | 67.1  | 66.9  | 60.3  | 56.0  | 48.7  | 44.5 |
|                                                                                  | 10%         | DB     | 39.8  | 42.2  | 50.6  | 63.9  | 72.2  | 76.7  | 80.0  | 79.7  | 68.4  | 58.2  | 48.8  | 44.1 |
|                                                                                  |             | MCWB   | 38.1  | 39.4  | 44.9  | 52.4  | 59.1  | 63.6  | 65.9  | 65.2  | 58.8  | 53.9  | 46.6  | 42.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 46.1  | 46.6  | 53.8  | 61.7  | 66.4  | 72.0  | 72.5  | 71.9  | 67.8  | 61.8  | 54.1  | 49.9 |
|                                                                                  |             | MCDB   | 47.8  | 48.9  | 62.0  | 76.7  | 78.9  | 84.8  | 82.8  | 84.7  | 83.5  | 69.0  | 56.5  | 51.2 |
|                                                                                  | 2%          | WB     | 43.4  | 44.0  | 50.9  | 58.6  | 64.5  | 69.7  | 70.4  | 69.8  | 65.2  | 58.8  | 51.0  | 47.2 |
|                                                                                  |             | MCDB   | 45.1  | 46.8  | 56.9  | 71.4  | 77.3  | 82.1  | 82.0  | 83.6  | 75.7  | 63.7  | 52.9  | 48.8 |
|                                                                                  | 5%          | WB     | 41.0  | 41.8  | 48.1  | 56.0  | 62.6  | 67.0  | 69.0  | 68.3  | 62.0  | 56.6  | 49.0  | 44.5 |
|                                                                                  |             | MCDB   | 42.8  | 44.5  | 54.0  | 65.8  | 74.0  | 77.7  | 80.0  | 80.3  | 69.0  | 61.0  | 50.7  | 46.2 |
|                                                                                  | 10%         | WB     | 38.1  | 39.5  | 45.5  | 53.4  | 60.6  | 64.9  | 67.5  | 66.7  | 60.0  | 54.4  | 46.8  | 42.0 |
|                                                                                  |             | MCDB   | 39.6  | 41.8  | 49.8  | 61.8  | 70.1  | 74.8  | 77.2  | 76.8  | 66.5  | 57.9  | 48.5  | 43.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.9   | 10.6  | 14.7  | 19.3  | 19.5  | 19.4  | 19.0  | 19.5  | 17.2  | 13.6  | 9.2   | 8.1  |
|                                                                                  |             | MCDBR  | 8.8   | 12.4  | 21.6  | 27.4  | 25.7  | 25.8  | 25.2  | 25.8  | 25.0  | 18.0  | 12.0  | 9.5  |
|                                                                                  |             | MCWBR  | 7.6   | 9.6   | 13.4  | 14.4  | 11.1  | 11.4  | 9.5   | 10.5  | 12.0  | 11.3  | 9.7   | 8.4  |
|                                                                                  | 5% WB       | MCDBR  | 8.7   | 11.8  | 19.3  | 24.3  | 22.8  | 22.2  | 21.5  | 22.3  | 20.9  | 16.4  | 11.6  | 9.4  |
|                                                                                  |             | MCWBR  | 7.6   | 9.6   | 12.7  | 13.4  | 11.3  | 12.3  | 9.5   | 10.2  | 11.8  | 11.1  | 9.8   | 8.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.312  | 0.341 | 0.374 | 0.420 | 0.415 | 0.408 | 0.440 | 0.429 | 0.387 | 0.366 | 0.346 | 0.309 |      |
|                                                                                  | taud        | 2.268  | 2.270 | 2.239 | 2.165 | 2.233 | 2.284 | 2.210 | 2.244 | 2.335 | 2.377 | 2.330 | 2.296 |      |
|                                                                                  | Ebn at Noon | 209    | 239   | 255   | 257   | 265   | 268   | 256   | 251   | 249   | 228   | 194   | 187   |      |
|                                                                                  | Edh at Noon | 20     | 27    | 35    | 42    | 41    | 40    | 42    | 39    | 31    | 25    | 19    | 17    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 203    | 420   | 811   | 1317  | 1648  | 1720  | 1587  | 1398  | 1003  | 548   | 228   | 147   |      |
|                                                                                  | RadStd      | 35     | 68    | 108   | 171   | 172   | 156   | 188   | 124   | 132   | 82    | 30    | 18    |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (30 neighbors) | +0.88 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -304  | +144  | N/A   |

Nomenclature: See separate page